

ABSTRACT ALGEBRA 1 - HOMEWORK 1- M. VICARÍA

Dear student please submit the complete solution to each of the following problems in Gradescope in a PDF file. Please write down your name in each page and specify clearly the problem that you stated. For your study purposes, I included a classification of the problems and their difficulties:

- Warm-up: these are require that you understand some definition.
- *: they indicate some medium level of difficulty. You should spend longer in these problems.
- ** indicates that the problem is challenging, they will require more time than the other problems.

The complexity from ** and * is the level we would like you to master in this course. Questions highlighted with **OPTIONAL** are not necessary to get full credit for the homework, but I highly encourage you to solve them for your learning process.

In each of the solutions you **SHOULD JUSTIFY COMPLETELY YOUR ANSWER** with a proof or a counter-example. No credit is granted to solutions without justification.

1. MANDATORY PROBLEMS

(1) **Problem 1(Functions):**

- (a) Let A, B and C three sets, and $f : A \mapsto B$ and $g : B \mapsto C$ be functions. We may compose them to get $g \circ f$. Show that if both f and g are injective so is $g \circ f$. Is the same true for surjectivity? If $g \circ f$ is surjective, what can be said about f and g ?
- (b) Let $f : S \mapsto T$ be a map. Prove that f is a bijection if and only if there is a map $g : T \mapsto S$ such that $g \circ f = Id_S$ and $f \circ g = Id_T$.

- (2) **Problem 2* : (Functions and equivalence relations)** Let f be a function from X to Y . Define the relation R on X by

$$xRy \text{ if and only if } f(x) = f(y).$$

Show that R is an equivalence relation. Describe the class $[x]$ for an element $x \in X$. In the case of the function $f : \mathbb{R}^2 \mapsto \mathbb{R}$ defined as (x, y) is sent to $x - y$, describe the equivalence classes geometrically.

(3) **Problem 3**:** (**Equivalence relations**) Let $X = \mathbb{R}^2 \setminus \{(0, 0)\}$. Let

$$E = \{((a_1, a_2), (b_1, b_2)) \in X \times X \mid (a_1, a_2) \text{ and } (b_1, b_2) \text{ lie in the same line that passes through the origin } (0, 0)\}$$

- (a) Prove that E is an equivalence relation on X .
- (b) Describe the class $[(1, 1)]$.
- (c) Give a geometric description of all classes of E .
- (d) Give a complete set of representatives of E .

2. OPTIONAL PROBLEMS

(1) **Optional Problem 1**:** $X = \mathbb{R} \times \mathbb{R} = \mathbb{R}^2$. Let

$$E = \{((a_1, a_2), (b_1, b_2)) \in \mathbb{R}^2 \times \mathbb{R}^2 \mid a_1 - b_1 \in \mathbb{Z} \text{ and } a_2 - b_2 \in \mathbb{Z}\}$$

- (a) Prove that E is an equivalence relation on X ,
- (b) Describe the class of $[(\pi, e)]$.
- (c) Let $r, s \in \mathbb{R}$, describe that class of $[(r, s)]$.

(2) **Optional Problem 2**:**

Recall: Let X be a set, then $\mathcal{P}(X)$ is the set of subsets of X .

Let X be a non-empty set and let $S \subseteq X$. Let $E_S = \{(A, B) \in \mathcal{P}(X) \times \mathcal{P}(X) \mid A \Delta B \subseteq S\}$, where $A \Delta B = A \cup B - (A \cap B)$.

- (a) Prove that E_S is an equivalence relation on $\mathcal{P}(X)$.
- (b) Let $S, T \subseteq X$ compare the following sets:
 - $E_S \cap E_T$ vs $E_{S \cap T}$,
 - $E_S \cup E_T$ vs $E_{S \cup T}$.